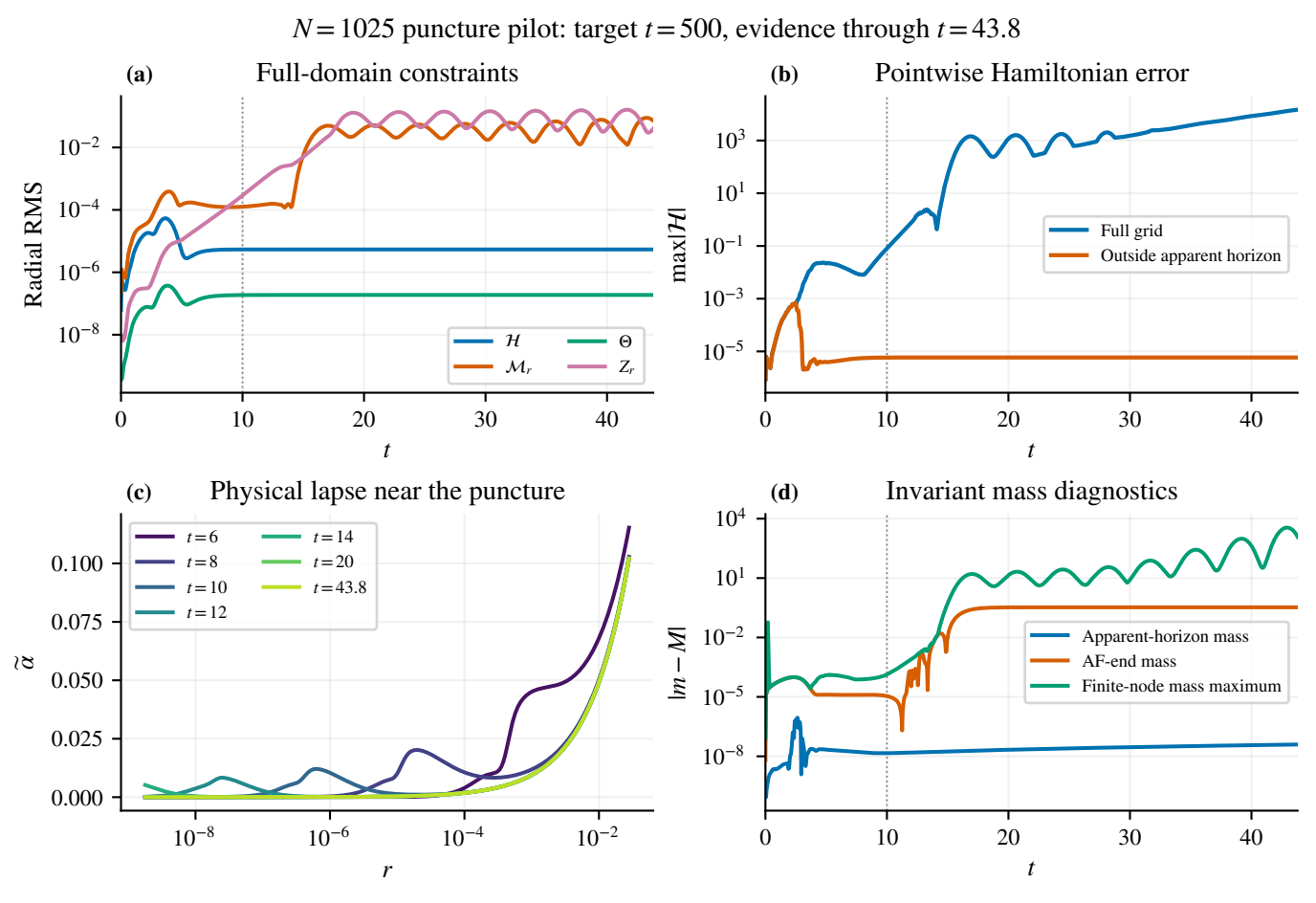
$N = 1025$ puncture pilot: target $t = 500$, evidence through $t = 43.8$
(a)
Full-domain constraints
Radial RMS
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$\mathcal{H}$
Θ
$\mathcal{M}_r$
Z_r
0
10
20
30
40
t
(b)
Pointwise Hamiltonian error
$\max|\mathcal{H}|$
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
Full grid
Outside apparent horizon
0
10
20
30
40
t
(c)
Physical lapse near the puncture
$\tilde{\alpha}$
0.100
0.075
0.050
0.025
0.000
$t = 6$
$t = 8$
$t = 10$
$t = 12$
$t = 14$
$t = 20$
$t = 43.8$
10^{-8}
10^{-6}
10^{-4}
10^{-2}
r
(d)
Invariant mass diagnostics
$|m - M|$
10^{4}
10^{1}
10^{-2}
10^{-5}
10^{-8}
Apparent-horizon mass
AF-end mass
Finite-node mass maximum
0
10
20
30
40
t